

VISHNU DHARAN K

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SUMMARY

AI Developer and Systems Architect specializing in autonomous multi-agent architectures, multimodal RAG retrieval systems, real-time voice AI, and production-grade full-stack LLM orchestration pipelines. B.Tech student in Artificial Intelligence and Data Science with 5+ AI projects spanning LangGraph agent orchestration, hybrid vector search, and low-latency speech pipelines, combined with strong full-stack engineering across FastAPI, React, and SQL/vector databases.

EDUCATION

Bannari Amman Institute of Technology

2023 – 2027

Bachelor of Technology in Artificial Intelligence and Data Science (CGPA-7.99)

Focused on AI system design, machine learning algorithms, database management systems, and web architecture through hands-on engineering projects.

SKILLS

AI & Generative AI: Prompt Engineering, RAG Systems, Multi-Agent Systems (LangGraph, LangChain), Voice AI

Languages: Python, Java, SQL, HTML5, CSS3

Backend & Orchestration: FastAPI, Flask, SQLAlchemy

Databases & Vector Engines: ChromaDB, MongoDB, MySQL, SQLite

Tools: Git, GitHub

PROJECT EXPERIENCE

AI-Powered Interactive Voice Lab Assistant

Voice AI & Speech Analytics · 2026

- Engineered a voice-enabled platform that automates lab viva evaluations, replacing manual, subjectively-scored assessments with real-time speech analytics.
- Combined Faster-Whisper for sub-second speech-to-text, Gemini 2.5 Flash for dynamic on-the-fly viva question generation, and Edge-TTS for audio synthesis.
- Built a JWT-authenticated React 19 educator dashboard surfacing scoring metrics, student logs, and performance trends.

Tech Stack: FastAPI, Gemini 2.5 Flash, Faster-Whisper, Edge-TTS, React 19, SQLAlchemy

Multi-Agent AI Recruitment & Talent Intelligence Platform

Autonomous Multi-Agent Architecture · HireHub Intelligence · 2025

- Built an multi-agent recruitment system to eliminate hundreds of manual hours spent screening resumes and scheduling interviews.
- Orchestrated multi-state agent workflows with LangGraph, handling resume parsing (PyMuPDF), semantic candidate matching, live AI interviews, and predictive scoring.
- Implemented high-precision semantic retrieval over a ChromaDB vector store, paired with a React + Vite recruiter interface for data-driven hiring recommendations.

Tech Stack: FastAPI, LangGraph, LangChain, ChromaDB, React + Vite, SpaCy

Jewellery Multimodal AI Retrieval System

Multimodal RAG Pipeline · 2025

- Architected a full-stack multimodal retrieval system enabling accurate jewellery discovery via text, image, or handwritten-tag search across 490+ inventory items.
- Combined CLIP visual embeddings, LLM-based query rewriting, and an OCR pipeline for handwritten tags to support flexible, natural search.
- Implemented hybrid FAISS (dense vector) + BM25 (sparse lexical) retrieval with cross-encoder reranking to maximize result precision.

Tech Stack: FastAPI, FAISS, CLIP, BM25, React + Vite, OpenCV

CERTIFICATIONS

Navigate Labs — Generative AI Program (On-Going)

In collaboration with Bannari Amman Institute of Technology · 2025 – 2026

10-month Generative AI Programme by Navigate Labs, co-certified with Bannari Amman Institute of Technology, covering intensive hands-on training across modern AI domains.